

# When Models Answer Without Seeing: Human-Calibrated Diagnosis of Linguistic Prior Exploitation in Vision-Language Models

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## Abstract

Analogous to hypothesis-only baselines that exposed annotation artifacts in Natural Language Inference (?), we investigate how much of VQA benchmark performance is achievable without any visual information. Across multiple open-source Vision-Language Models (VLMs), blind inference with a blank image still yields substantial VQA accuracy, often exceeding human blind performance, indicating strong exploitation of linguistic priors shaped by the training distribution (?). To characterize this behavior, we introduce *control question variants*: four progressively weaker reformulations of each question that remove lexical anchors while preserving question type. We further analyze how blind model behavior changes under a single additional instruction, revealing a spectrum from *instruction-gated* prior use to more unconditional hallucination. Blind outputs are also highly overconfident: for several models, 80–90% of outputs have logprob  $> -0.5$ , a calibration failure invisible to accuracy-only evaluations. Our human dataset (N=24, 641 questions, IRB-approved) provides a reference point for blind reasoning without corpus-frequency bias. Preliminary alignment experiments suggest that human blind responses may provide a useful supervision signal, though robust cross-validated estimates remain future work.

## 1 Introduction

Vision-Language Models (VLMs) achieve impressive performance on visual question answering (VQA) benchmarks (??), yet a basic question remains unanswered: how much of this performance reflects genuine visual grounding, and how much can be explained by linguistic priors inherited from large-scale language model pretraining?

This question parallels a well-known episode in Natural Language Inference (NLI): ? showed that a model trained on hypotheses alone—ignoring the corresponding premises—substantially exceeded the random baseline across ten NLI datasets. The finding revealed annotation artifacts embedded in the benchmarks themselves. We apply the same diagnostic logic to VQA: the *image* plays the role of the premise, and the question is the hypothesis. A model that answers correctly from the question alone is exploiting a hypothesis-only shortcut.

? provide the mechanistic explanation: large language models are shaped by their training objective over internet-scale text, and therefore produce outputs proportional to

the probability of sequences in the training corpus. When a VQA question has a high-frequency answer in training data (“What color is the sky?”  $\rightarrow$  “blue”), a VLM will produce that answer reliably regardless of the image. Blind VQA accuracy thus directly measures corpus frequency exploitation.

In this work, we conduct a diagnostic study of this phenomenon across multiple open-source VLMs. Our study has four interlocking components. First, we evaluate models under *blind* conditions (blank image input) on 1,000 VQA questions, finding that several models substantially outperform human participants answering the same questions without images. Second, we introduce *control question variants*—four progressively weaker reformulations that strip away lexical anchors—to measure how robustly models exploit priors as linguistic content is removed. Third, we analyze model *confidence* via logprobabilities, revealing severe overconfidence in hallucinated answers and a systematic confidence boost when models are given permission to use their priors. Fourth, we compare all model behaviors against a human reference dataset (N=24 participants, 641 questions) collected under the same blind conditions, where humans are not subject to corpus-frequency bias and thus serve as a calibration baseline for what blind inference *should* look like.

### Main findings.

1. VLMs outperform humans on blind VQA (avg.  $\sim 48\%$  vs.  $39\%$  on VQA), with significantly different answer distributions, confirming corpus-frequency exploitation rather than genuine reasoning.
2. Models exhibit characteristic null-hypothesis biases under blind conditions: 70% of yes/no answers are “no”, 75% of count answers are “0”, and color answers cluster on “black”. These biases partially reverse under instruction, revealing the prior activation mechanism.
3. We identify an *instruction-gated* vs. *unconditional* hallucination spectrum: Qwen3-VL-8B converts 72% of hedging responses into specific answers when given an instruction to imagine the scene, while LLaVA v1.5 converts only 17%—hallucinating unconditionally regardless of permission.
4. Models are severely overconfident: 80–90% of blind outputs have logprob  $> -0.5$ , and the instruction further

raises confidence by  $\sim 0.08$  log-probability points uniformly.

5. Initial fine-tuning experiments suggest that human blind responses may improve human-model alignment without obvious loss of visually-grounded accuracy, but the current evidence is still based on single-split results.

## 2 Related Work

**Hypothesis-Only and Shortcut Baselines.** ? demonstrated that NLI classifiers achieve non-trivial accuracy using only the hypothesis, exposing annotation artifacts. ? further characterized these artifacts through systematic analysis of annotation bias. In VQA, ? showed that models relying on question-answer co-occurrence statistics could achieve high accuracy without visual understanding. Our work extends this diagnostic tradition by providing a human reference dataset alongside model evaluations, enabling calibrated measurement of the gap.

**LMs as Corpus Frequency Models.** ? showed that LLM failure patterns are predicted by three factors: the probability of the task, the input, and the output in the training corpus. Models are biased toward high-probability outputs regardless of task requirements. We operationalize this insight: blind VQA accuracy directly measures the probability of question-answer pairs in the pretraining distribution.

**VLM Hallucination and Shortcut Learning.** Hallucination in VLMs—confident generation of visually ungrounded content—has been widely documented (???). ? proposed MMStar to test genuine visual grounding, requiring fine-grained and logical reasoning. VLind-bench (?) specifically measures language prior influence across diverse benchmarks. ? identified visual shortcuts through spurious image features. We complement these by providing a *behavioral* characterization—how and when models commit to prior-driven answers.

**Human Cognition Under Uncertainty.** Human reasoning under missing information is characterized by fast, heuristic System 1 thinking (?): people rely on representativeness, availability, and anchoring. Importantly, humans do not share the corpus-frequency bias of pretrained LLMs, making them a natural reference point for what linguistically-driven blind inference should look like in the absence of training-distribution shortcuts.

## 3 Experimental Setup

### 3.1 Models and Datasets

We evaluate open-source VLMs from the Qwen3-VL (?) and LLaVA (??) families, plus Qwen3 (?) LLMs as text-only baselines. The main blind-performance table reports the model subset for which the current zero-shot results are fully assembled; additional instruction-only comparisons are reported separately in Appendix D. All models are evaluated in zero-shot settings with temperature 0.

[TODO: Add matched decoder-only baselines using the same inference pipeline but without image input. Preferred backbone matches: Qwen3-VL  $\rightarrow$  Qwen/Qwen3-8B; LLaVA-Vicuna  $\rightarrow$

lmsys/vicuna-7b-v1.5; LLaVA-Mistral  $\rightarrow$  mistralai/Mistral-7B-Instruct-v0.2.

If the current code does not cleanly support `image=None`, add a decoder-only branch rather than forcing a blank image. Also decide whether to drop InternVL entirely due to Titan RTX limits or replace it with a lighter supported model such as Qwen/Qwen2.5-VL-7B-Instruct.]

Our primary evaluation dataset consists of 1,000 questions sampled from the VQA v2.0 validation split (?), spanning yes/no, numerical, and open-ended question types. We additionally evaluate on 267 questions from the Science & Technology category of MMStar (?), which targets visual reasoning skills including coarse perception (CP), fine-grained perception (FP), instance reasoning (IR), and logical reasoning (LR). For the human study and fine-tuning analysis (Section 6), we use a 641-question subset consisting of 374 VQA and 267 MMStar questions.

### 3.2 Blind Inference Conditions

We evaluate models under three conditions: (1) **Standard**: the original image with the question; (2) **Blind**: a blank  $224 \times 224$  white PNG replacing the original image; and (3) **Blind+Inst**: blank image with an additional instruction appended to the question: “*Note: No images are provided. For each question, imagine an appropriate image exists and answer based on the most common or universal scenario.*”

### 3.3 Control Question Variants

To characterize *how* models exploit linguistic priors, we introduce four progressively weaker reformulations of each question, generated using GPT-4.1-mini (temperature 0, seed 42, JSON schema mode) for reproducibility:

1. **deictic\_removed**: Deictic markers (*this, that, here, there*) and spatial qualifiers (left/right/top/bottom) are removed while preserving fluency. Tests whether spatial anchoring drives shortcut behavior.
2. **object\_removed**: The specific object noun is replaced with the neutral placeholder “object.” Tests the contribution of entity identity to prior exploitation.
3. **weaker\_object**: The specific noun is replaced with a one-level-up hypernym (dog  $\rightarrow$  animal, car  $\rightarrow$  vehicle, shirt  $\rightarrow$  clothing). Tests whether category-level identity is sufficient to trigger prior recall.
4. **pronominalized**: All content noun phrases are replaced with minimal pronouns following grammar-aware rules (“What color is the dog?”  $\rightarrow$  “What color is it?”; “How many cats are there?”  $\rightarrow$  “How many are there?”). This maximally removes entity-specific prior cues while preserving question structure.

The degradation chain from original  $\rightarrow$  pronominalized forms a monotonically decreasing axis of lexical prior cues, enabling a controlled measurement of how much each entity reference contributes to shortcut exploitation. The GPT extraction prompt and JSON schema are provided in Appendix C.

### 3.4 Human Data Collection

We recruited  $N=24$  participants (9 men, 15 women; age 20–43) through online university community forums. All procedures were conducted under an IRB-approved protocol (compensation: 10,000 KRW/hour, consistent with standard rates). Participants answered 641 VQA-style questions in a single session ( $\approx 1$  hour) using the same blank-image blind condition as the models, relying only on the text of the question and general world knowledge. Responses could be given in Korean or English and were translated and normalized prior to analysis. After each answer, participants rated their confidence on a 5-point Likert scale. Full protocol details are provided in Appendix A.

### 3.5 Metrics

We report: (1) **VQA accuracy** using the standard 10-annotator soft evaluation protocol (?); (2) **semantic similarity** via Sentence-BERT (?) cosine similarity for open-ended answers; (3) **Multimodal Gain** ( $MG = S_{\text{vision+text}} - S_{\text{text-only}}$ ), following ?, quantifying the performance change when visual input is added; (4) **Spearman  $\rho$**  between model and human per-question correctness as human alignment score; and (5) **logprobability confidence**, extracted as the mean token-level log-probability of the generated answer (top-5 logprobs per token, temperature 0); (6) **Pearson difficulty correlation**: Pearson  $r$  between the human group accuracy curve  $h(q)$  (proportion of 20 participants correct per question) and per-model accuracy, measuring alignment at the question-difficulty level rather than the individual rater level; and (7) **Krippendorff’s  $\alpha$**  (nominal scale) for within-group and cross-group inter-rater reliability, which handles multiple raters simultaneously and avoids the prevalence-inflation issue of Cohen’s  $\kappa$  and Gwet’s AC1. Note:  $h(q)$  is computed from VQA soft-scores (0/0.33/0.67/1.0 averaged across raters); model accuracy is binary (0 or 1). This asymmetry is inherent to the task structure and is consistent with standard VQA evaluation.

## 4 Linguistic Prior Exploitation at Scale

### 4.1 Blind Accuracy Across Models

Table 1 reports zero-shot blind performance for all models and humans across VQA and MMStar. Several patterns emerge.

**VLMs outperform humans on blind VQA.** The best models in the current comparison achieve  $\sim 48$ – $49\%$  average VQA accuracy under blind conditions, compared to humans at  $39.0\%$ . This gap is especially pronounced on yes/no questions (VLM:  $68$ – $75\%$  vs. human:  $66.6\%$ ) and open-ended questions (VLM:  $23$ – $29\%$  vs. human:  $13.0\%$ ).

**LLMs perform comparably to VLMs on blind VQA.** Qwen3 LLMs (text-only backbones, no vision encoder) achieve  $35.3$ – $38.8\%$  on VQA, close to their VLM counterparts. This confirms that blind performance is driven by the language component rather than the vision encoder.

**Scale effects differ by dataset.** Within the currently tabulated model set, larger models do not consistently improve blind VQA accuracy, consistent with the ceiling on high-frequency question-answer patterns in the training distribu-

*Regenerate from ACL Figure 1 with updated 1K-question data.  
Top: yes/no distribution (Models vs Humans vs GT).  
Bottom: count distribution (Models vs Humans vs GT).  
Key values: model “no” = 70%, human “no” = 53%, GT = 47%;  
model count “0” = 75%, human count is distributed.  
Source: evaluation/logs/pretrained/ blind files.*

Figure 1: Response distributions for yes/no (top) and numerical (bottom) questions comparing models, humans, and ground truth under blind conditions. Models exhibit a strong “no” default (70%) and a “zero” default (75% of count answers), whereas humans show more calibrated distributions. [TODO: Regenerate from 1K blind logits.]

tion. On MMStar, some larger variants show stronger logical reasoning (LR), suggesting that this benchmark may benefit more from scale.

**Models fail to recognize the absence of visual input.** Adding the blind+inst instruction produces only modest accuracy changes in many settings, though not uniformly below 2pp on every model (see Table 2), indicating that model blind accuracy is not strongly modulated by awareness of the missing image. Instead, models generate plausible answers regardless.

### 4.2 Answer Distribution Biases

A key question is not just *whether* models answer correctly under blind conditions, but *how* their answer distributions differ from humans and from ground truth. As shown in Figure 1, three systematic biases emerge:

**No-bias in yes/no questions.** Under blind conditions, 70% of model yes/no answers are “no” (vs. ground truth: 47% no). Humans show a smaller bias (53% no), suggesting that the model’s extreme no-bias reflects an epistemic default—the null answer under uncertainty—rather than a corpus prior. Notably, this reverses under blind+inst (“yes” increases from 30% to 55%), confirming that the “no” default is instruction-suppressed rather than a genuine linguistic prior.

**Zero-bias in count questions.** 75% of model numeric answers under blind conditions are “0,” while humans produce a much more distributed count distribution. The interpretation is consistent: in the absence of visual evidence, models default to “there are none.”

**Black-bias in color questions.** Among open-ended color answers, “black” accounts for 16% of model responses under blind conditions, far exceeding its ground-truth frequency. This likely reflects the modal color in the training distribution for generic object descriptions.

These null-hypothesis biases illustrate that models are not randomly sampling from their prior—they produce a specific failure mode that partially mimics the structure of a reasonable uncertainty response.

Table 1: Zero-shot blind performance of humans and models across datasets. VQA “Other” metric: Accuracy/Similarity. MMStar subcategories: CP (coarse perception), FP (fine-grained perception), IR (instance reasoning), LR (logical reasoning). Bold: best; underline: second best.

| Model              | LLM     | Size | VQA  |      |                  |             | MMStar      |             |             |             |             |
|--------------------|---------|------|------|------|------------------|-------------|-------------|-------------|-------------|-------------|-------------|
|                    |         |      | Y/N  | Num  | Other            | Avg         | CP          | FP          | IR          | LR          | Avg         |
| Humans (N=24)      | –       | –    | 66.6 | 13.8 | 13.0/45.5        | 39.0        | 28.2        | 26.3        | 28.4        | 27.7        | 27.4        |
| LLM Baseline       |         |      |      |      |                  |             |             |             |             |             |             |
| Qwen3              | Qwen3   | 4B   | 54.9 | 19.0 | 24.7/55.2        | 38.8        | 25.0        | 33.7        | 23.6        | 40.9        | 29.7        |
|                    |         | 8B   | 50.3 | 17.1 | 22.1/53.2        | 35.3        | 17.6        | 29.7        | 20.0        | <u>59.1</u> | 26.8        |
| Multi-Modal Models |         |      |      |      |                  |             |             |             |             |             |             |
| Qwen3-VL           | Qwen3   | 2B   | 71.8 | 17.1 | 23.8/55.5        | 46.4        | 30.9        | 28.7        | 21.8        | 22.7        | 27.2        |
|                    |         | 4B   | 71.5 | 19.0 | <u>28.9/56.4</u> | <u>48.0</u> | 17.6        | <b>34.7</b> | 25.5        | 40.9        | 28.5        |
|                    |         | 8B   | 71.5 | 21.9 | 27.0/54.9        | <b>48.0</b> | 30.9        | 30.7        | <b>30.9</b> | 50.0        | <u>32.5</u> |
| LLaVA v1.5         | Vicuna  | 7B   | 70.3 | 17.1 | 19.4/48.4        | 43.8        | <b>36.8</b> | 27.7        | <u>29.1</u> | 27.3        | <b>30.5</b> |
| LLaVA v1.6         | Mistral | 7B   | 69.6 | 20.0 | 27.0/55.0        | 47.0        | 25.0        | 24.8        | 21.8        | 31.8        | 24.8        |
| LLaVA v1.6         | Vicuna  | 7B   | 74.4 | 20.0 | 27.2/56.2        | <u>49.4</u> | 32.4        | 20.8        | <u>30.9</u> | 18.2        | 26.0        |

Regenerate from ACL Figure 2 (MG bar chart) with updated models.

X-axis: Shared Wrong — Human-Only — Shared Correct — Model-Only.

Y-axis: Mean MG. Blue = VQA, Orange = MMStar.

Source: analysis/notebook/analysis-quadrants.ipynb.

Figure 2: Multimodal Gain (MG) by correctness quadrant for VQA (blue) and MMStar (orange). MG is highest when both humans and models fail blindly (Shared Wrong, Human-Only), and collapses for questions models already answer correctly via priors (Shared Correct, Model-Only). [TODO: Regenerate from analysis-quadrants.ipynb.]

### 4.3 Multimodal Gain Analysis

To assess when visual information actually helps, we compute Multimodal Gain (MG) across four correctness quadrants: questions where both human and model are wrong (**Shared Wrong**), where only the human is correct (**Human-Only**), both correct (**Shared Correct**), or only the model (**Model-Only**). Figure 2 shows mean MG per regime.

MG is largest in the **Shared Wrong** and **Human-Only** quadrants, meaning visual information provides the greatest benefit precisely for questions where blind inference fails for both humans and models. Conversely, in the **Shared Correct** and **Model-Only** quadrants, MG collapses near zero or turns negative, indicating that visual information *degrades* performance for questions that models already answer correctly via linguistic priors. This suggests that adding vision can interfere with an already-active prior, a form of distributional conflict between visual evidence and prior belief.

[TODO: Degradation curve: accuracy + confidence across  
original → deictic\_removed → object\_removed →  
weaker\_object → pronominalized, one line per model.  
Requires unified 10-annotator scoring. See  
data\_issues.md.]

Figure 3: Control question degradation curves: accuracy (solid) and mean logprob confidence (dashed) across the five control variant types, for each model family. [TODO: Fill after scoring fix.]

## 5 Characterizing Prior Reliance

### 5.1 Control Question Degradation

To quantify how much each lexical element drives prior exploitation, we measure model accuracy and confidence across the four control variants defined in Section 3.3. However, the current control-variant analysis is not yet stable enough to support strong quantitative claims: scoring harmonization and missing control outputs remain unresolved for part of the model set.

This subsection therefore serves primarily to define the evaluation protocol. We defer interpretation of degradation curves until the scoring regime is fully harmonized across the original and control questions.

### 5.2 Instruction-Gated Hallucination

A key question is whether models’ blind answers reflect a readily available prior that is always active, or a latent prior

[TODO: Bar chart: soft abstention collapse rate per model.

Qwen3-VL-8B=71.9%, LLaVA-Mistral=51.9%,  
LLaVA-Vicuna=28.3%, LLaVA-1.5=17.1%.

X-axis: model; Y-axis: % collapse; grouped by  
architecture family.]

Figure 4: Soft abstention collapse rate: the fraction of hedging blind responses (“nothing”, “none”, “nowhere”) that convert to specific answers under the Blind+Inst condition. The spectrum reveals an instruction-gated (Qwen3-VL) vs. unconditional (LLaVA v1.5) hallucination axis. [TODO: Generate figure from analysis numbers.]

that requires explicit permission to express. We investigate this by comparing model outputs between the **Blind** and **Blind+Inst** conditions.

We classify model outputs in the blind condition as: *hard-abstained* (explicit refusal: “I cannot answer”, “no image”); *soft-abstained* (hedging outputs: “nothing”, “none”, “unanswerable”, “nowhere”); *hallucinated\_correct* (specific answer matching GT); or *hallucinated\_wrong* (specific answer not matching GT). Hard abstentions are near-zero across all models (0.0–0.3%), consistent with the baked-in “Answer using a single word” instruction suppressing explicit refusals. The real signal is *soft abstention*: 11–19% of blind outputs are hedging responses, varying by model family.

Figure 4 shows the *soft abstention collapse rate*: the fraction of soft-abstained blind responses that convert to specific answers under the Blind+Inst instruction. This rate varies dramatically:

| Model              | Soft-abs (blind) | Collapse rate |
|--------------------|------------------|---------------|
| Qwen3-VL-8B        | 12.9%            | 71.9%         |
| LLaVA v1.6-Mistral | 15.8%            | 51.9%         |
| LLaVA v1.6-Vicuna  | 11.3%            | 28.3%         |
| LLaVA v1.5         | 18.7%            | 17.1%         |

We interpret this spectrum as measuring the degree to which prior activation is *instruction-gated*. Qwen3-VL, a highly instruction-tuned model, largely suppresses its prior under blind conditions but rapidly activates it when given permission. LLaVA v1.5, by contrast, hallucinates *unconditionally*: it produces specific answers whether or not instructed to, and shows almost no response to the permission instruction. This distinction is mechanistically important: instruction-gated models have a functioning “uncertainty” signal that instruction can override, while unconditional models lack this gate entirely.

Additionally, 33.6% of all answers (not just soft-abstentions) change between blind and blind+inst conditions for Qwen3-VL-8B—a large behavioral shift driven by a sin-

gle sentence. The yes/no distribution shifts from 70% “no” in blind to 55% “yes” in blind+inst, consistent with the instruction activating a “plausible scenario exists” prior that overrides the default epistemic null.

### 5.3 Confidence Calibration

Logprobability analysis reveals that blind hallucinations are produced with extreme confidence. For Qwen3-VL-8B under blind conditions, 80.8% of output tokens have mean logprob  $> -0.5$  ( $\approx 60\%$  probability per token), and the mean logprob across all blind outputs is  $-0.258$ . Under Blind+Inst, mean logprob rises to  $-0.177$  and the fraction with logprob  $> -0.5$  increases to 89.5%. The instruction does not merely change *which* answer is produced—it also makes the model more *certain* about whatever it produces.

The few soft-abstained outputs (“nothing”, “none”) have lower mean logprob ( $-0.37$  vs.  $-0.26$  for answered outputs), confirming that hedging outputs reflect genuine uncertainty. However, the effect size is small, and the baseline level of confidence even for soft-abstentions is high. The primary calibration failure is not that models are uncertain when they should be confident, but that they are *highly confident about wrong hallucinated answers* they should be uncertain about.

[TODO: Compute Expected Calibration Error (ECE) and plot reliability diagrams (confidence bins vs. accuracy) for blind vs. standard conditions across models. Expected finding: blind condition shows severe overconfidence, high ECE; standard condition better calibrated. See confidence.ipynb.]

[TODO: Run corpus frequency correlation: correlate per-question blind accuracy with answer frequency in VQA v2 training split. Expected:  $r > 0.5$ , directly operationalizing the McCoy (2024) mechanism.]

## 6 Human-Model Alignment

### 6.1 Humans as a Calibration Reference

Human blind performance provides a natural reference point for what linguistically-driven inference *should* look like in the absence of corpus-frequency shortcuts. Unlike VLMs, human participants answering without images must rely on general world knowledge and explicit reasoning—they do not have a training-distribution prior over question-answer pairs.

This distinction is empirically visible in answer distributions: while models produce a bimodal yes/no distribution dominated by “no” (70%), humans show a more calibrated distribution (53% no) that better tracks ground-truth frequencies (47% no). For numerical answers, 75% of model answers are “0” while human count distributions are spread across small integers, consistent with humans reasoning about plausible scenarios rather than defaulting to absence.

Human accuracy on VQA is 39.0% averaged across question types. Humans perform below the evaluated models on numerical questions as well (13.8% vs. 17–22% for models), and models substantially outperform humans on yes/no and other answer types. On MMStar, both humans and mod-

els perform near random ( $\approx 27\%$ ), confirming that MMStar successfully targets visual grounding.

## 6.2 Human-Model Divergence in Answer Patterns

We measure human-model alignment using two complementary metrics. First, we compute **Pearson difficulty correlation**: for each question  $q$ , we compute the human group accuracy  $h(q)$  (proportion of 20 participants correct) and correlate it with per-model accuracy. We prefer Pearson over Spearman here because the relationship is approximately linear (max  $|r - \rho| = 0.026$  across all conditions), and operating on proportions avoids the information loss from rank conversion on binary data. Second, we compute **Krippendorff’s  $\alpha$**  (nominal scale, binary correctness) to measure within-group and cross-group reliability.

Human difficulty estimates are highly reliable:  $\text{ICC}(2, 20) = 0.963$  across 20 raters, justifying use of  $h(q)$  as a stable difficulty signal. Within-group human reliability is  $\alpha_{\text{human}} = 0.554$  (moderate, appropriate for a genuinely ambiguous task without visual input).

Pretrained VLMs show moderate correlation with human difficulty under blind conditions (Pearson  $r = 0.50$ – $0.62$  depending on model), confirming that human and model difficulty patterns partially overlap—both exploit linguistic priors to some extent. However, adding any model to the human rater pool *decreases*  $\alpha$  (from 0.554 to 0.535–0.543), confirming that models are structurally less consistent with the human group than humans are with each other.

In the current analysis, Blind+Inst often yields slightly higher human-difficulty correlations than Blind alone for several model families. This pattern is consistent with the soft-abstention-collapse result in Section 5.2, but it should be treated as provisional until the human comparison conditions are fully matched across settings.

The inter-rater agreement matrix (Figure 5) further illustrates the structural separation: human participants cluster together in terms of question-level consistency patterns, while models form a separate cluster despite operating under identical conditions. This separation is not merely about accuracy level—it reflects qualitatively different reasoning strategies.

Semantic similarity analysis reinforces this finding: model answers have higher embedding similarity to ground truth (mean 65.8 vs. human 64.2) despite being generated without visual evidence, and the human similarity distribution has a longer left tail (more diverse wrong answers), consistent with exploratory reasoning rather than prior retrieval.

## 6.3 Fine-tuning to Human Prior Distributions

We fine-tune VLMs on human blind response distributions to test whether the human-model alignment gap can be closed without sacrificing visual grounding. We apply LoRA adapters (rank 32, scale 64) to all linear layers while freezing the vision encoder and alignment modules, training with AdamW ( $\text{lr} = 2 \times 10^{-5}$ , cosine schedule, 5% warmup, 5 epochs) on  $N=15$  participant annotations.

We investigate two objectives: (1) **SFT**: standard cross-entropy on the most frequent human canonical answer; and

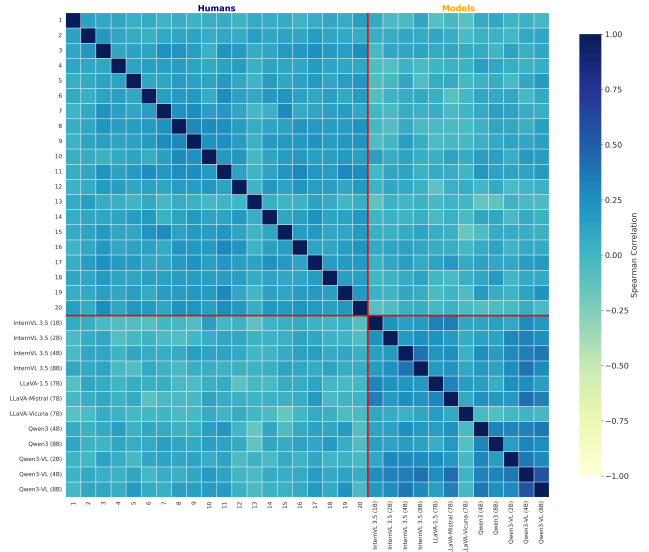


Figure 5: Inter-rater agreement matrix under blind conditions (Spearman  $\rho$ , pairwise). Human participants (upper block) cluster separately from models (lower block), indicating structurally different question-difficulty patterns despite identical task conditions.

(2) **JS**: distributional alignment via Jensen-Shannon divergence loss (?) between model and human confidence distributions over canonical answers.

Initial single-split results suggest alignment improvement: Spearman  $\rho$  increases from 0.455 (pretrained) to 0.598 (SFT) and 0.547 (JS), while visual accuracy on the standard VQA task appears broadly preserved. We treat these numbers as preliminary until the planned k-fold evaluation is complete.

**[TODO: Report k-fold (5-fold) mean  $\pm$  std  $\rho$  values to replace single-split results. Infrastructure in `train_kfold_sft.py` and `dataset/folds/`. If InternVL is removed due hardware limits, update this section and figures to use only runnable model families. Table format: `model  $\times$  objective  $\times$  mean  $\rho \pm$  std.]`**

Figure 6 shows alignment ( $\rho$ ) vs. visual accuracy for pretrained, SFT-aligned, and JS-aligned models. Fine-tuned models improve  $\rho$  while maintaining accuracy on the visual diagonal, whereas fine-tuning on ground truth (blue markers) fails to improve blind alignment—confirming that the alignment target (human blind responses) matters, not merely fine-tuning itself.

## 6.4 Model Analysis-Guided Human Study

**[TODO: Design and run new human study on model-analysis-sampled questions.]**

The analyses in Sections 5–6 reveal specific question types for which model behavior is most diagnostic: questions where (a) models are highly confident but wrong (**Tier A**); (b) answers change across control variants (**Tier B**); (c) all models give the same wrong answer (**Tier C**); and (d) accuracy drops sharply from original to pronominalized (**Tier**

*Regenerate from ACL Figure 3 (alignment scatter) with k-fold error bars.*  
*X-axis: Spearman  $\rho$  (0.40–0.65). Y-axis: VQA accuracy (74–90%).*  
*Markers: circle=Pretrained, square=SFT, triangle=JS.*  
*Blue=GT-trained, Orange=Human-blind-trained. Red dashed = human baseline  $\rho$ .*  
*Add: error bars from 5-fold CV. Add: Qwen3-VL results.*  
*Source: analysis/notebook/analysis-finetuned.ipynb.*

Figure 6: Human-model alignment ( $\rho$ , x-axis) vs. standard VQA accuracy (y-axis) for pretrained (circles), SFT-aligned (squares), and JS-aligned (triangles) models. Blue markers use ground-truth training; orange markers use human blind responses. Fine-tuning on human blind responses (orange) improves  $\rho$  while maintaining accuracy; fine-tuning on ground truth (blue) does not improve alignment. [TODO: Regenerate from analysis-finetuned.ipynb with k-fold error bars.]

D).

We plan a targeted human study on a sample of 300 questions drawn from these tiers (Section F), where participants answer each question in all five control variant forms. This enables two new analyses not possible with the original study: (i) human degradation curves comparable to Figure 3, testing whether humans also degrade monotonically as lexical content is removed or remain stable by relying on structural cues; and (ii) direct comparison of human and model prior sensitivity to entity removal.

## 7 Discussion

**The instruction-gated spectrum as an architectural insight.** The dramatic variation in soft-abstention collapse rates (17% for LLaVA v1.5 vs. 72% for Qwen3-VL) suggests that instruction-following capability plays a role in prior activation that goes beyond accuracy. Instruction-tuned models have learned to gate prior expression behind contextual permission signals, while models with weaker instruction following leak priors unconditionally. This is not simply a capability difference—LLaVA v1.5 is competitive on standard VQA accuracy—but a behavioral difference in how priors are accessed.

**Implications for benchmark design.** Our results suggest that any VQA benchmark can be partially solved by linguistic priors alone, and that the magnitude of this effect varies predictably with question type, answer distribution, and entity specificity. We propose that blind VQA evaluation should become a standard diagnostic for new VQA benchmarks: a high blind accuracy (>40%) is at least a warning sign that the benchmark may over-index on knowledge-accessible question types.

**Limitations.** Our human study involves 24 participants recruited from university communities in a single language (Korean/English), limiting cultural and demographic generalizability. Model evaluation uses greedy decoding (temperature 0), which may underestimate answer diversity and overestimate confidence. The control variant GPT extraction uses a best-effort seed (seed is advisory in the Responses API) and results may vary slightly across runs. The corpus

frequency correlation analysis is pending and represents an important gap in our mechanistic account.

## 8 Conclusion

We present a human-calibrated diagnostic study of linguistic prior exploitation in Vision-Language Models. Analogous to hypothesis-only baselines in NLI, blind VQA evaluation reveals that VLMs can answer many questions without visual input, often by exploiting corpus-frequency priors rather than image evidence. We introduce control question variants as a way to probe this reliance, and use confidence analysis to expose the severe overconfidence that accompanies blind hallucination. The instruction-gated hallucination finding reveals a useful behavioral spectrum across model families. Human blind responses provide a reference point for evaluating these behaviors and a promising, though still preliminary, target for alignment.

### A Human Study Protocol

A total of 24 participants (9 men and 15 women; age range: 20–43 years) were included in the study. Participants were recruited via online student community forums at universities. All procedures were conducted under an IRB-approved protocol and adhered to the principles of the Declaration of Helsinki. Informed consent was obtained from all participants prior to participation.

Each participant completed 641 VQA-style questions in a single session, which took approximately one hour on average. Participants were compensated at a rate of 10,000 KRW per hour, consistent with standard compensation for similar annotation tasks.

For VQA, participants were instructed to provide free-form answers that were as concise as possible. Responses could be given in either Korean or English and were translated and processed using the same answer normalization and aggregation strategy described in Appendix B. MMStar questions were presented with the original multiple-choice options, along with Korean translations, in the same order as the dataset.

## B Answer Aggregation

We cluster semantically equivalent answers (e.g., “3” and “three”) into canonical forms using GPT-4o mini (?). The prompt instructs the model to group synonymous responses, select the most concise canonical form, and return results in structured JSON. Human responses are then aggregated into a confidence-weighted distribution over canonical answers, where each answer’s weight is the sum of confidence scores from all annotators selecting that answer, normalized to sum to 1:

$$H(a_k^*) = \frac{\sum_{i: \phi(a_i) = a_k^*} c_i}{\sum_i c_i} \quad (1)$$

where  $c_i$  is participant  $i$ ’s Likert confidence (normalized to  $[0, 1]$ ),  $\phi$  maps raw answers to canonical answers, and  $K$  is the number of canonical answer clusters.

## C Control Question Extraction Prompt

Control variants are extracted using GPT-4.1-mini-2025-04-14 (temperature 0, seed 42) with strict JSON schema mode for reproducibility. The system prompt instructs the model to generate all four variants in a single API call per batch of 50 questions. The schema enforces: `{deictic_removed, object_removed, weaker_object, pronominalized}` as required string fields, with `additionalProperties: false`. For pronominalized, the prompt specifies grammar-aware substitution rules to ensure natural pronominalization (“What color is it?” not “What color is the object?”).

## D Blind Awareness Instruction

After the original prompt and the blank image, models in the Blind+Inst condition receive: “*Note: No images are provided. For each question, imagine an appropriate image exists and answer based on the most common or universal scenario.*”

Table 2 shows accuracy comparison between Blind and Blind+Inst conditions. The instruction usually changes accuracy by only a few points, confirming that it does not reliably improve performance even when it substantially changes answer behavior.

Table 2: Accuracy comparison with and without blind instruction.  $\Delta_{\text{inst}} = \text{Blind}_{\text{inst}} - \text{Blind}$ .

| Model          | VQA  |       |          | MMStar |       |          |
|----------------|------|-------|----------|--------|-------|----------|
|                | GT   | Blind | $\Delta$ | GT     | Blind | $\Delta$ |
| InternVL3.5-1B | 80.5 | 43.6  | 0.2      | 43.4   | 25.7  | -1.9     |
| InternVL3.5-2B | 82.1 | 45.5  | -0.2     | 51.7   | 26.1  | 0.7      |
| InternVL3.5-4B | 83.0 | 45.0  | 2.2      | 64.3   | 24.5  | -2.9     |
| InternVL3.5-8B | 86.9 | 47.1  | -1.7     | 62.7   | 28.7  | -0.8     |
| Qwen3-VL-2B    | 83.0 | 45.0  | 0.7      | 51.7   | 23.9  | 3.3      |
| Qwen3-VL-4B    | 85.9 | 45.5  | 1.9      | 63.8   | 21.8  | -3.0     |
| Qwen3-VL-8B    | 89.4 | 44.3  | 3.1      | 66.0   | 25.2  | 1.8      |

## E Training Details

We apply LoRA to all linear layers with rank 32 and scaling factor 64, while freezing the vision encoder and alignment modules. We optimize using AdamW with a learning rate of  $2 \times 10^{-5}$ , cosine learning-rate scheduling with 5% warmup, and weight decay of 0.1, training for a maximum of 5 epochs. We select the checkpoint with the best validation performance for evaluation. Hyperparameters were selected based on prior literature and preliminary runs rather than exhaustive optimization.

## F Question Sampling for New Human Study

To select maximally diagnostic questions for the new human study (Section 6.4), we stratify 1,000 control questions into five tiers based on model behavior:

- **Tier A:** High confidence + wrong (mean logprob  $> -0.25$ , accuracy  $< 20\%$ ). Questions where models hallucinate with extreme certainty.
- **Tier B:** Answer changes across control variants ( $\geq 2$  answer changes across 4 variants). Questions sensitive to linguistic surface form.
- **Tier C:**  $\geq 80\%$  of models wrong on baseline. Shared corpus prior (hard questions with misleading priors).
- **Tier D:** Large accuracy drop from original to pronominalized ( $\Delta \text{acc} > 30\text{pp}$ ). Shortcut-dependent questions where entity name drives the answer.
- **Tier E:** Large confidence drop from original to pronominalized ( $\Delta \text{logprob} > 0.2$ ). Questions where entity removal creates uncertainty even if the answer is unchanged.

The target sample is 300 questions:  $\sim 60$  from each of Tiers A/B/D/E and  $\sim 120$  random (stratified by entity type, capped at 20% per type).